PRISM: See The Full Spectrum

A Multi-Perspective Platform for Democratic Resilience

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[TPINIA7005] Understanding Emerging Technologies

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I. Introduction

The digital information ecosystem faces an unprecedented crisis. While misinformation has existed throughout history, the current environment differs fundamentally. The information now spreads faster than verification. Algorithmic curation creates self-reinforcing echo chambers, and it collapses the traditional editorial gatekeeping that has enabled unverified claims to reach millions within hours (Pariser, 2011; Bakshy et al., 2015). Political polarization has intensified as individuals increasingly consume news from ideologically homogeneous sources, making cross-partisan dialogue nearly impossible (Bruns, 2019). This challenge extends beyond individual media choices. It threatens the foundation of democratic discourse itself.

Recent developments have exacerbated these challenges. Artificial intelligence systems can now generate convincing but entirely fabricated content at scale, from deepfake videos of political figures to synthetic news articles indistinguishable from human-written journalism (Hill & Freedman, 2025; Hsu & Lu, 2024). Traditional fact-checking organizations struggle to keep pace with the volume of misinformation. The attempts to label content as "true" or "false" often backfire, which reinforces existing biases rather than changing minds.

PRISM (Political Reality Information Spectrum Mapper) addresses this challenge through a fundamentally different approach. Rather than attempting to be an arbiter of truth, which is an impossible task in an era where "facts" themselves are contested, PRISM reconstructs the complete information picture surrounding political events and socially significant issues. While misinformation affects many domains, PRISM initially targets political news and politically-charged social issues, where the polarization and echo chambers are most severe. Our political spectrum framework naturally applies to debates around healthcare, climate policy, immigration, and economic issues. This allows us users to have comprehensive multi-perspective coverage. As the platform matures, we envision expanding our classification systems to other domains.

Our team brings together expertise in policy analysis, user experience design, data science, and social media research. This interdisciplinary approach enables us to address misinformation not merely as a technical problem but as a sociotechnical challenge requiring careful consideration of human behavior, institutional incentives, and democratic values. The core innovation of PRISM lies in its recognition that truth in complex political matters is multi-faceted, not binary. Instead of telling users what to believe, we show them what they're missing. Our platform organizes diverse sources across the political spectrum, provides credibility indicators, tracks how information evolves over time, and empowers users to draw their own informed conclusions.

II. Literature Review & Competitor Landscape

A. The Problem: Echo Chambers and Algorithmic Polarization

The concept of "filter bubbles" is about personalized information environments that isolate users from contrary viewpoints. It has become central to understanding modern media consumption (Pariser, 2011). Bakshy et al. (2015) found that while social media platforms technically expose users to diverse content, algorithmic curation and selective engagement result in significantly reduced cross-ideological information exposure. Bruns (2019) extends

this analysis, demonstrating that filter bubbles operate not merely through algorithmic selection but through the entire sociotechnical assemblage of platform design, user behavior, and content moderation.

Notably, the rise of generative AI has introduced new dimensions to this crisis. Hill and Freedman (2025) document how chatbots can enter "delusional spirals," generating increasingly disconnected and false information while maintaining an authoritative tone. Hsu and Lu (2024) illustrate the danger of synthetic media through cases like AI-generated celebrity endorsements that deceive thousands before detection. These technological developments compound existing challenges, making the need for multi-perspective information systems increasingly urgent.

On the other hand, research on media literacy interventions offers some hope. Guess et al. (2020) demonstrated that even brief digital literacy training can significantly reduce the sharing of false news online. However, such interventions require users to actively engage with fact-checking processes, which is a barrier that limits scalability. PRISM builds on this insight by embedding diverse perspectives directly into the information consumption experience, reducing the friction required for critical thinking.

B. Competitor Analysis and Differentiation

Several platforms address aspects of media polarization, but each has significant limitations:

1) **AllSides** curates news by political bias but organizes content by topic rather than specific events, making it difficult to see how different outlets cover the same developing story. Users must manually navigate between left, center, and right sections, creating friction that reduces engagement.

2) **Ground News** provides multi-perspective news coverage with visual indicators of political bias. While valuable, it lacks event-based organization and fact-checking integration, functioning primarily as a news aggregator rather than an analytical tool for understanding information ecosystems.

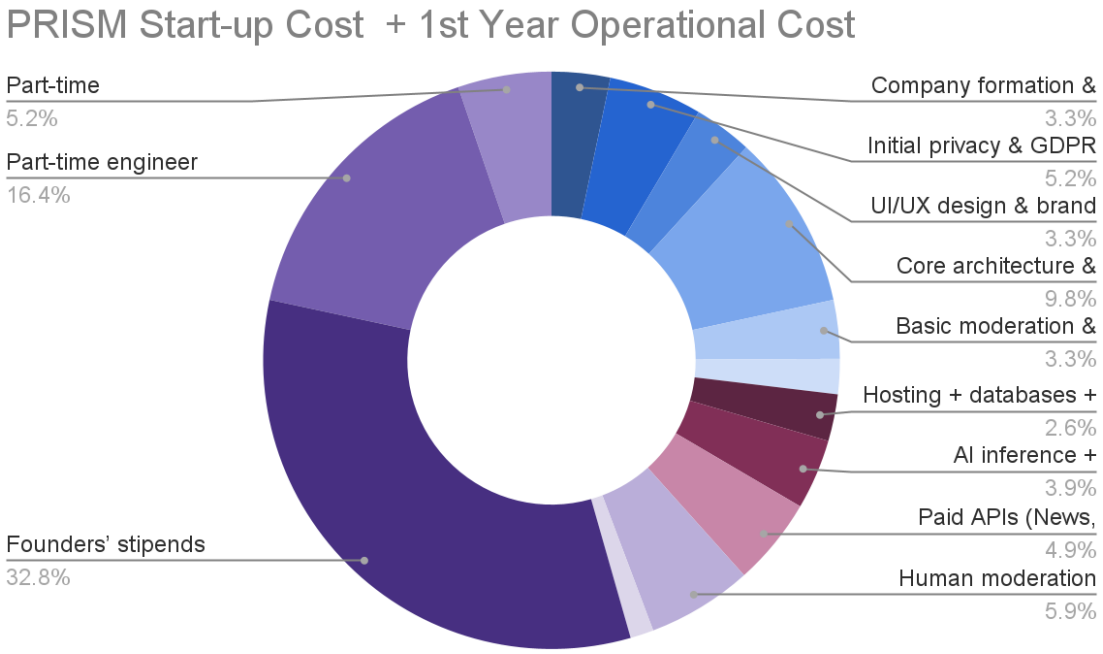
3) **Snopes and PolitiFact** offer rigorous fact-checking but operate reactively, which only addressing claims after they've spread widely. Fundamentally, their binary "true/false" verdicts can be counterproductive in polarized environments, where audiences dismiss fact-checks that contradict their existing beliefs.

4) **Google News** aggregates content from diverse sources but relies heavily on algorithmic personalization, which potentially reinforces filter bubbles rather than breaking them. It provides no systematic political spectrum analysis or credibility indicators.

PRISM's differentiation lies in its integrated approach: (1) event-centric organization with chapter-based timelines, (2) simultaneous political spectrum and credibility visualization, (3) proactive information reconstruction rather than reactive fact-checking, (4) community-driven inquiry marketplace, and (5) personal information archive for curated knowledge building.

III. Finances

PRISM's business model has three priorities to ensure our product is: (1) viable, (2) accountable, and (3) scalable. During the first year, our budget covers both setup costs and operational costs. While the setup costs aim to lay out the groundwork for future operations (e.g., company formation, platform development, legal compliance), the operational costs ascertain smooth operations of the business. We estimate that the former will cost us $26,000-$41,000 and the latter $75,000-$121,000, totaling $101,000-$162,000 for the first year.



Our business is expected to be financed through various sources of revenues that align with the platform's mission of improving media literacy and understanding complex information ecosystems. Rather than relying on invasive advertising or data monetization, PRISM emphasizes subscription access, institutional partnerships, and mission-aligned grants, reflecting its commitment to ethical technology and user privacy. Our three main revenue streams include:

1) **Freemium Subscription Model:** PRISM offers a free tier that provides access to core features such as multi-perspective views and basic information analytics, while reserving more advanced features (e.g., full trend history, creator dashboards, export tools, and an ad-free experience) for premium subscribers. This model is intended to convert approximately 5-10% of users into paying customers. If PRISM reaches ten thousand users in the first year, between five hundred and one thousand premium subscribers could generate several thousand dollars per month. A similar tiered model is offered for creators, enabling emerging journalists or content producers to access additional analytics, audience demographics, and monetization features.

2) **Institutional Licensing:** This represents a second major revenue source with strong potential for growth. Universities, colleges, and schools can license PRISM to support media literacy education and research, while enterprises can adopt PRISM as part of employee digital literacy or responsible media engagement programs. Educational licenses are priced per student per semester, and enterprise pricing scales with

organization size. These offer a predictable and sustainable revenue pathway. We can also expect revenues to come from research access, where academic users can access the platform free of charge while commercial research institutions pay for API access and historical datasets. This aligns with PRISM's role as a research-grade platform for journalism, information science, and political communication studies.

3) **Non-Profit and Foundation Support:** We can partner with journalism and democracy-focused organizations and ethical tech foundations, including but not limited to the Knight Foundation, Mozilla Foundation, Open Society Foundations, and Omidyar Network. Their grants contribution can provide significant early-stage funding. These grants directly support the social and educational value PRISM delivers, while reducing reliance on commercial monetization during early development.

Altogether, PRISM anticipates generating between $165,000 and $230,000 in the first year, which would be sufficient to cover operational costs and approach sustainability even during the MVP stage. As subscriptions, institutional partnerships, and research access expand in years two and three, PRISM projects full sustainability and potential profitability. Importantly, this multi-stream model allows PRISM to grow ethically without compromising user privacy or autonomy, ensuring long-term alignment between financial sustainability and social mission.

Total Addressable Market (TAM)

Our primary audience consists of information-conscious professionals (ages 25-45), university students engaged in civic issues (ages 18-25), and educators seeking media literacy tools (ages 30-55). In the United States alone, we estimate approximately 10 million information-conscious professionals and 4 million actively engaged university students represent our addressable market. Our realistic three-year target is 100,000 active users, with potential expansion to international markets as the platform matures.

IV. Design

A. Core Product Philosophy

PRISM operates on a fundamental principle: complex events have no single truth, but rather multiple valid perspectives shaped by different values, evidence interpretations, and framing choices. Our goal is not to tell users what to believe but to show them the full spectrum of perspectives they might otherwise miss. This approach addresses misinformation not by labeling content as "true" or "false", which often backfires in polarized environments. Instead, PRISM makes information gaps visible and provides tools for users to navigate complexity.

B. Key Features

Political Spectrum Visualization

Our core differentiator is the interactive political spectrum mapper that positions news sources on a continuous scale from progressive (-10) to conservative (+10). Major outlets are pre-mapped based on established media bias research, with sources ranging from The Nation and MSNBC on the progressive end to Fox News and Daily Wire on the conservative end, with Reuters and Associated Press near center. Each source includes: political lean score (numerical and visual), credibility rating (0-100, based on fact-checking track record), source type (Legacy

Media, Digital Native, Expert Analysis, Official Statement), and framing description (how this source typically covers this topic). Users can filter by spectrum position, seeing only center sources, only opposing viewpoints, or the full range. This granular control empowers users to actively engage with diverse perspectives rather than passively receiving algorithmically curated feeds (see Figure 1).

Event-Based Timeline Organization

Unlike traditional news aggregators that present streams of unrelated articles, PRISM organizes information around specific unfolding events with hierarchical chapter structures. For example, a major political event might be divided into chapters tracking the initial incident, investigation phase, official response, and ongoing policy debate. This structure helps users understand how events develop over time and how different outlets emphasize different phases of a story (see Figure 2).

Integrated Fact-Checking

PRISM incorporates fact-checking directly into event pages, displaying verified claims with color-coded verdicts (TRUE, MOSTLY TRUE, PARTIALLY FALSE, FALSE, UNVERIFIED). Each fact-check includes explanations and sources, allowing users to understand the reasoning behind verdicts. Importantly, PRISM acknowledges the limits of fact-checking: we clearly distinguish between objective factual claims (which can be verified) and subjective interpretations or value judgments (which cannot).

Information Journey Tracking

PRISM visualizes how information spreads over time, showing a timeline of coverage, reach progression (Local → National → Global), source diversity, and viral patterns. This meta-information helps users understand not just what happened, but how the public learned about it and which narratives gained traction.

Community Q&A Marketplace

Users can submit questions about events, upvote questions from others, and track investigation status (OPEN / INVESTIGATING / ANSWERED). This creates a demand signal for journalistic attention, potentially connecting professional fact-checkers and investigative journalists with genuine public information needs (see Figure 3).

Personal Information Archive

Users can bookmark events of interest, building a curated knowledge base over time. This archive serves multiple purposes including personal reference for complex ongoing stories, pattern recognition across different events, and contribution to anonymized research data on information consumption patterns.

C. Prototype Implementation

We have developed a working web-based prototype demonstrating PRISM's core functionality using a real-world case study: the December 2024 UnitedHealthcare CEO shooting. This event exemplifies the type of politically-charged story PRISM is designed to handle, sparking intense debate about healthcare policy, corporate accountability, and societal violence, which belongs to topics that different outlets framed through distinctly different lenses. The prototype successfully demonstrates that PRISM's approach is technically feasible and that users can quickly grasp the multi-perspective framework. The interface features interactive political

spectrum visualization with sources positioned across the ideological range, event timeline with chapter-based organization, integrated fact-checking for specific claims, and mobile-responsive design optimized for smartphone use.

V. Conclusion

Misinformation and political polarization represent existential challenges to democratic societies. Traditional approaches, such as fact-checking after the fact, algorithmic content moderation, media literacy education, all are valuable but face fundamental limitations. PRISM offers a path that is not focusing on presuming to know the truth, but making information diversity accessible, visible, and engaging. By organizing events across the political spectrum, integrating credibility indicators, and empowering users to explore multiple perspectives, we address the root cause of filter bubbles which is not the existence of biased sources, but the invisibility of alternative viewpoints.

Our financial model demonstrates viability through diversified revenue streams that align with our mission. Our design balances user empowerment with practical usability. Our team brings the interdisciplinary expertise necessary to navigate the complex intersection of technology, media, and democracy. Altogether, our pitch offers an innovative emerging technology solution towards news consumption in the current digital and social media landscape. As Angwin et al. (2016) demonstrated in their investigation of algorithmic bias, systems shape behavior in ways their designers often don't anticipate. PRISM is designed with this awareness that we do not claim to eliminate bias or misinformation, but rather to make the information ecosystem more transparent, more diverse, and more conducive to informed citizenship.

References

Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016, May 13). Machine bias. *ProPublica*. https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

Bakshy, E., Messing, S., & Adamic, L. A. (2015). Exposure to ideologically diverse news and opinion on Facebook. *Science, 348*(6239), 1130–1132. https://doi.org/10.1126/science.aaa1160

Bruns, A. (2019). Filter bubble. *Internet Policy Review, 8*(4). https://doi.org/10.14763/2019.4.1426

Guess, A. M., Lerner, M., Lyons, B., Montgomery, J. M., Nyhan, B., Reifler, J., & Sircar, N. (2020). A digital media literacy intervention increases discernment between mainstream and false news in the United States and India. *Proceedings of the National Academy of Sciences, 117*(27), 15536–15545. https://doi.org/10.1073/pnas.1920498117

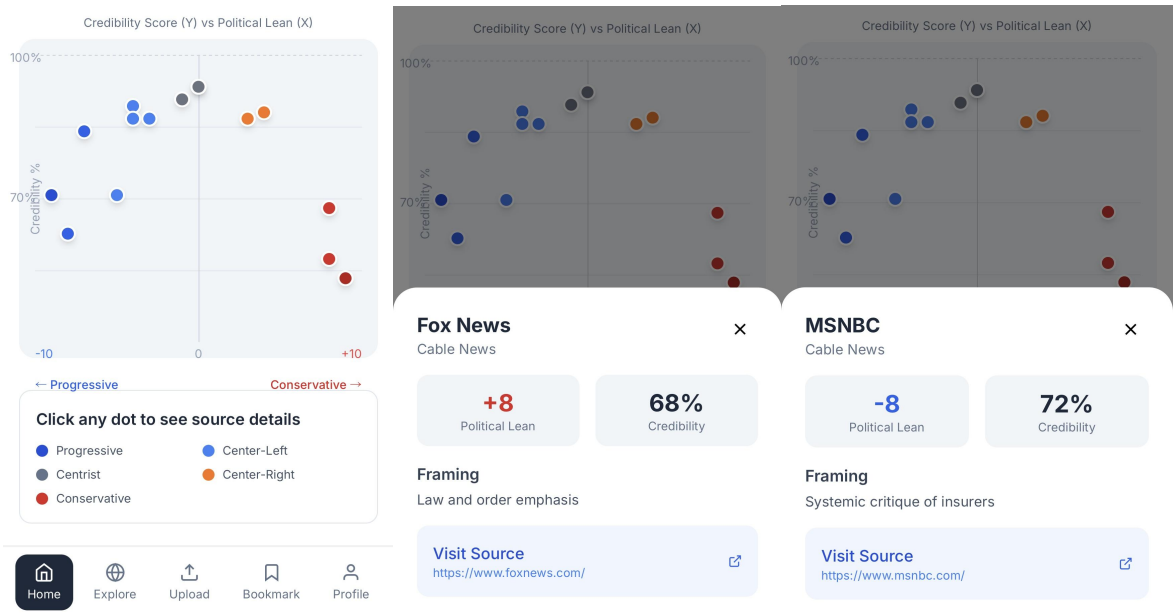
Hill, K., & Freedman, D. (2025, August 8). Chatbots can go into a delusional spiral. Here's how it happens. *The New York Times*. https://www.nytimes.com/2025/08/08/technology/ai-chatbots-delusions-chatgpt.html

Hsu, T., & Lu, Y. (2024, January 9). No, that's not Taylor Swift peddling Le Creuset cookware. *The New York Times*. https://www.nytimes.com/2024/01/09/technology/taylor-swift-le-creuset-ai-deepfake.html

Pariser, E. (2011). *The filter bubble: What the Internet is hiding from you*. Penguin Press.

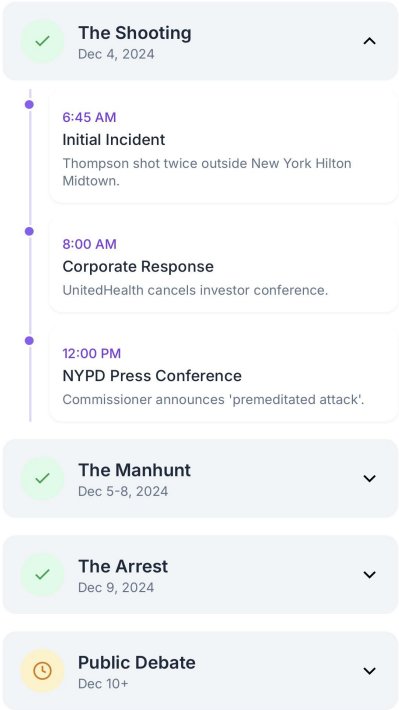
Appendix: Product Interface Mockups

Figure 1: Political Spectrum Visualization Interface



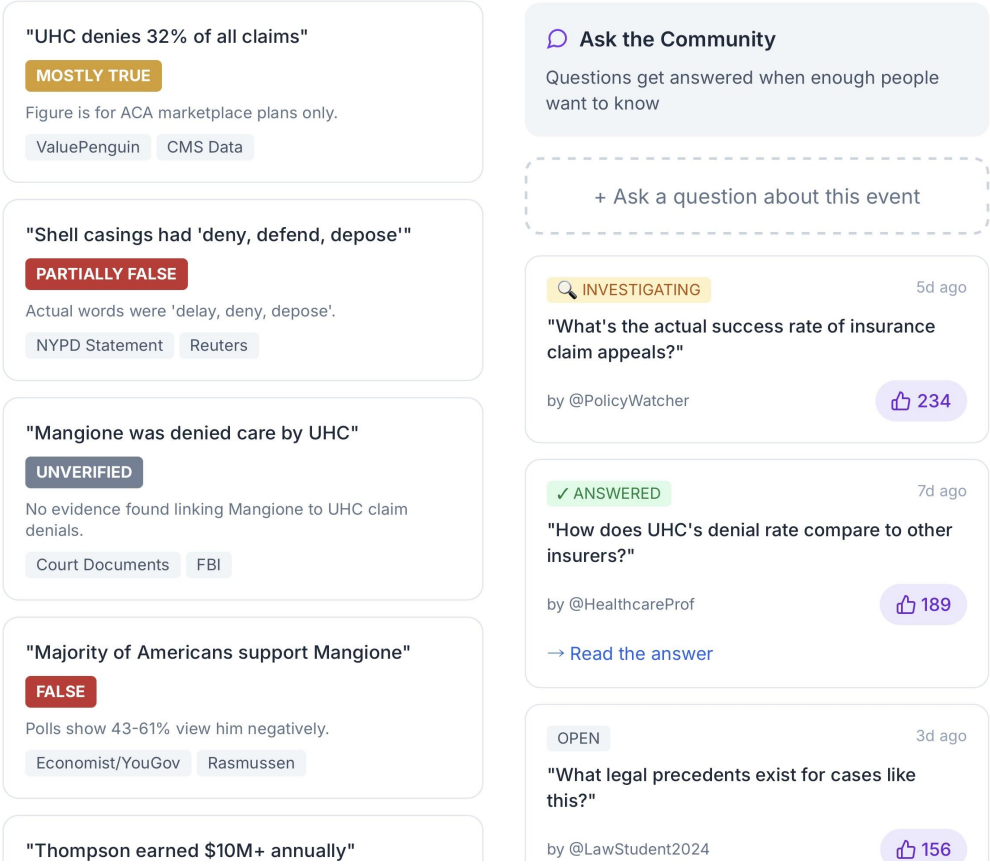
Interactive political spectrum mapper displaying news sources positioned from progressive (-10) to conservative (+10). Each source includes credibility scores, source types (Legacy Media, Digital Native, Expert Analysis, Official Statement), and framing descriptions. Example event shows UnitedHealthcare CEO shooting coverage across 15 major outlets, demonstrating how PRISM allows users to compare coverage from The Nation and MSNBC (progressive) to Fox News and Daily Wire (conservative), with Reuters and Associated Press positioned near center.

Figure 2: Event Timeline with Chapter Organization



Hierarchical event structure with expandable chapters demonstrating PRISM's event-based organization: (1) The Shooting, (2) The Manhunt, (3) The Arrest, (4) Public Debate. Each chapter displays timestamp, resolution status (resolved/ongoing), and the number of sources covering that specific phase. This visualization helps users understand how events develop over time and how different media outlets emphasize different aspects of the narrative.

Figure 3: Community Q&A and Fact-Checking Interface



Integrated fact-checking panel featuring color-coded verdicts (TRUE in green, MOSTLY TRUE in light green, PARTIALLY FALSE in yellow, FALSE in red, UNVERIFIED in gray) alongside community-submitted questions. Each fact-check includes detailed explanations and source citations. Community questions display upvote counts and investigation status (OPEN / INVESTIGATING / ANSWERED), creating a marketplace for journalistic attention where users can collectively identify information gaps and track progress toward answers.

Prototype: https://tpi-emerging-tech.netlify.app/

A functional web-based prototype demonstrating PRISM's core features using the UnitedHealthcare CEO shooting case study. The prototype includes interactive political spectrum visualization, event timeline navigation, integrated fact-checking, and mobile-responsive design optimized for smartphone use.